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Potential and economic viability of standalone hybrid systems for a rural community of Sokoto, North-west Nigeria

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Abstract An assessment of the potential and economic viability of standalone hybrid systems for an off-grid rural community of Sokoto, North-west Nigeria was conducted. A specific electric load profile was developed to suite the community consisting 200 homes, a school and a community health center. The data obtained from the Nigeria Meteorological Department, Oshodi, Lagos (daily mean wind speeds, and daily global solar radiation for 24 years from 1987 to 2010) were used. An assessment of the design that will optimally meet the daily load demand with a loss of load probability (LOLP) of 0.01 was performed, considering 3 stand-alone applications of photovoltaic (PV), wind and diesel, and 3 hybrid designs of wind-PV, wind-diesel, and solar-diesel. The diesel standalone system (DSS) was taken as the basis of comparison as the experimental location has no connection to a distribution network. The HOMER® software optimizing tool was engaged following the feasibility analysis with the RETScreen software. The wind standalone system (WSS) was found to be the optimal means of producing renewable electricity in terms of life cycle cost as well as levelised cost of producing energy at \$0.15/(kW·h). This is competitive with grid electricity, which is presently at a cost of approximately \$0.09/(kW·h) and 410% better than the conventional DSS at a levelised cost of energy (LCOE) of \$0.62/kWh. The WSS is proposed for communities around the study site.

Keywords photovoltaic (PV) power, wind power, solar-wind hybrid, cost per kilowatt-hour, clean energy

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1 Introduction

Access to sustainable, dependable and competitively priced energy anchors almost every aspect of life in any modern economy [1]. The essential factors that alter future energy systems are demographics, urbanization, incomes, demand for energy and market liberalization [2]. The availability of energy resources is one of the factors that have been identified as having the potential of introducing deep seated changes to the energy sector. Other factors include new technologies, such as solar photovoltaic (PV), and social and personal priorities [2,3]. The social and personal priorities have been focused to create a steady shift from high to low carbon fuels, driven by social requirements for cleaner and more accessible energy.

Presently the widely used energy generation modules across several countries are those of fossil fuel and nuclear power origin. These sources have proved harmful both to the environment and to humans. Based on this, there are campaigns to either totally shift or diversify the energy generation modules to renewable energy (RE) resources for power generation. The reason for this is that they have been found to be environmentally friendly, widely available, easily applicable and non toxic [4–6].

Worthy of note is the fact that improvements in standards of living are manifested in increased food production, increased industrial output, the provision of efficient transportation, adequate shelter, healthcare and other human services [7]. All these are connected to an increase in energy consumption [8]. Therefore, there are losses in the economy when modern power sources for production activities are unavailable. An estimate by the Council for Renewable Energy of Nigeria places the losses due to power outages at about 126 billion naira (US\$ 984.38 million) annually [9]. Based on this, several industries and households depended on self generation through the use of fossil-fuel-based generators. In 1985, the Central Bank of Nigeria estimated that the nation consumed approximately 8771863 tons of crude oil equivalents [10]. The value corresponds to about 180000

barrels of crude oil daily. This figure has been constantly increasing ever since. Furthermore, the Department for Petroleum Resources [11] reported that 78% of the total amount of energy consumed in Nigeria is petroleum based. The byproduct of such practice is deleterious to the environment and humans. Thus one of the recognized and accepted ways of meeting development without destruction of the environment is to adopt RE sources for power generation [12,13].

2 Potential of renewable energy resources in Nigeria

Many indigenous researchers have looked into the potential of RE resources in Nigeria with a view to establishing their viability in the country. Onyebuchi [10] projected the technical potential of solar energy in Nigeria using a 5% conversion efficiency apparatus. The outcome led to a realization of 15.0×10^{14} kJ of useful energy per annum. This amount of energy equates to about 258.62 million barrels of oil yearly [14] and corresponds to 30% of the 876 million barrels of annual crude oil production in the country as of February, 2013. This also amounts to about 4.2×10^5 GW·h of yearly electricity production, which is about 26 times the recent annual electricity production in the country [15]. A work by Chineke and Igwiro [16] shows that Nigeria receives abundant solar energy that can be profitably exploited with an annual average daily solar radiation of about $5.25 \text{ kW}\cdot\text{h}/(\text{m}^2\cdot\text{d})$, with variations between $3.5 \text{ kW}\cdot\text{h}/(\text{m}^2\cdot\text{d})$, along the coastal areas of the south and $7.0 \text{ kW}\cdot\text{h}/(\text{m}^2\cdot\text{d})$ at the north. The average duration of sunshine hours all over the country was estimated at 6.5 h with an average annual solar energy intensity of $1935 \text{ kW}\cdot\text{h}/(\text{m}^2\cdot\text{a})$ —this approximates to a solar energy of $1770 \text{ TW}\cdot\text{h/a}$ falling on the entire Nigeria land mass. This approximately equals a multiple of 120000 of the total annual average electrical energy generated by the Power Holding Company of Nigeria (PHCN) [17]. Hence the retrievable solar energy with a 10% unadventurous conversion efficiency yields about 23 times the energy demand projection of the Energy Commission of Nigeria for the year 2030 [18]. Therefore, it is necessary to include solar energy in the nation's energy mix.

On the opportunities for wind-to-electricity projects in Nigeria, a number of research reports exist. For instance Adekoya and Adewale [14] analyzed wind speed data of 30 stations in Nigeria and found the annual mean wind speeds and power flux densities to vary from 1.5 to 4.1 m/s and $5.7\text{--}22.5 \text{ W}/\text{m}^2$, respectively. Another research by Fagbenle and Karayiannis [19] on a 10-year wind data from 1979 to 1988, considered surface and higher winds as well as maximum gusts. Ngala et al. [18] conducted statistical and cost benefit analyses of the wind energy

potential of a site in Maiduguri employing Weibull statistics on a 10-year wind data spanning 1995 to 2004. Ajayi [20] hinted that inland, the wind is greatest in hilly regions of the North, while upland topographies of the middle belt and northern edges of the nation have enormous potential for huge wind energy harvest. Mean wind speeds in the north and south were reported to lie from 4.0 to 7.5 m/s and 3.0 to 3.5 m/s respectively at 10 m height [14]. It was the conclusion of many researchers that wind energy is particularly of significant abundance at the core northern states, the hilly and mountainous parts of the central and eastern states, and also the country's offshore areas [5,14,21–23]. They all pointed to the fact that the nation is blessed with a vast natural supply of solar and wind energy, with huge opportunity for electricity production [24]. Despite this, however, the energy need of citizens in the rural areas is still hinged on traditional biomass [24]. This group of fuels makes up more than 50% of total energy usage in Nigeria [25]. More so, fuel wood supply/demand imbalance in some parts of the country is becoming a risk to the energy security of the rural communities [26–31], because of the extent to which deforestation has taken place. Therefore, with a very low annual per capita consumption of electricity estimated between $100 \text{ kW}\cdot\text{h}$ and $135 \text{ kW}\cdot\text{h}$ [32] and the knowledge that over 100 local governments areas of Nigeria are not connected to the national electricity grid [14], a diversification of the nation's energy mix would suffice. Thus knowing that RE resources have the advantage of being employed for standalone facility in addition to grid connectivity, employing such will help the country achieve the target of energy security by the year 2020 [33].

3 Present work

Few research studies exist that have assessed the potential of hybrid RE system for power generation in Nigeria. Nwosu et al. [34] explored the prospect of generating electricity either as grid-connected or as standalone hybrid power system. The hybrid was designed to suit the load demand of a 3-bedroom flat. It was observed that the hybrid plant was capable of satisfying the daily average load demand of about 1.5 kW in the hot season and less than 1.0 kW in the rainy season. Mbakwe et al. [35] proposed a standalone PV-wind hybrid energy system (HES) with battery storage for a cellular mobile telecommunications base station site in a remote location in Benin City and found out that a hybrid system with diesel generator backup appeared to be cost effective. Abatcha et al. [36] designed and simulated a hybrid power system that targeted remote users. The main power for the hybrid system was to come from the PV panels, while the fuel cell (FC) and secondary batteries were used as backup units. In addition, a life cycle cost (LCC) analysis of a diesel/PV

hybrid power generating system for an off-grid residential building in Enugu, Nigeria was also carried out by Agajelu et al. [37], which focused on optimizing different hybrid system configurations, and contrasting the end results with the PV standalone (PVSA) and conventional diesel standalone (DSA) systems. The result showed that the hybrid system had an internal rate of return that was 1.7% higher than that of the PV standalone system.

Reports on analysis of the feasibility, economic viability and design of hybrid systems that can provide sustainable energy for rural communities are still at a pioneering phase. The few existing works are based on small scale generation for remote telecom applications and individual buildings, as discussed above. This study is, therefore, focused on this. It aims to analyze and determine the viability of utilizing RE hybrid system for rural communities in Sokoto (13°01'N, 5°15'E), North-west, Nigeria. Part of the study is focused on the determination of the adequate turbine parameters for the sites, and the site specific PV sizing. Other focuses include the optimal sizing of the hybrid systems for this location and the cost benefit analysis of the individual generation as compared to the hybrid generation.

4 Materials and method

4.1 Data collection

The 20-four year (1987–2010) daily global solar radiation and daily wind speed data employed in this study were sourced from the Nigeria Meteorological agency (NIMET), Oshodi, Lagos, Nigeria. The study focused on designing a standalone hybrid system for a rural community in Sokoto, with 200 homes, having a school and a community health center. The location view of the selected site is shown in Fig. 1. Two 25 kW wind turbines, cumulative solar panels of 135 kW and a diesel generator of 35 kW were employed for the study as standalone or hybrid power systems.



Fig. 1 Location of meteorology station of the community (Google Earth)

4.2 Load calculation

The load profiles for rural communities unconnected to the grid cannot be accessed from the database of the electric utility company, though electricity usage among rural families in developing nations is quite low, (at an average of 1 kW·h/d per home [38]) due to several factors such as unavailability of expensive appliances and high rate of poverty. The statistics for electricity consumption in India in 2009 was found to be 96 kW·h annually per capita in rural areas [39]. This equates to about 315 kW·h per day for the studied community of 200 homes in Sokoto. The energy demand requirement of the rural community was, therefore, evaluated by analyzing the individual power rating of the appliances utilized in each home. Therefore, the assumed average electricity consumption value used for this study is 1.4 kW·h per day per home, while the mode of analysis is presented in Tables 1 and 2 [40–42]. Figure 2 presents the hourly load profile of 24 h for the community.

Table 1 General wattage chart for some household appliances

Power rating/W	Household appliance
24	42" ceiling fan (low speed)
55–90	19" CRT television
150–340	Desktop computer and 17" CRT monitor
60	60-W light bulb (incandescent)
18	CFL light bulb (60-W equivalent)

4.3 Modeling the PV project

The RETScreen® solar radiation model employed for the study was as proposed by Klein and Theilacker according to Duffie and Beckman [43]. However, the model was broadened to include the case of moving surfaces, while the PV array model engaged was based on the work by Evans [44]. The concept of daily utilizability [43] was employed to determine the portion of the load that can be met directly by the PV array. The energy balance analysis evaluates the portion of the load required to be met by the battery and diesel generator, if present.

4.3.1 Description of the algorithm

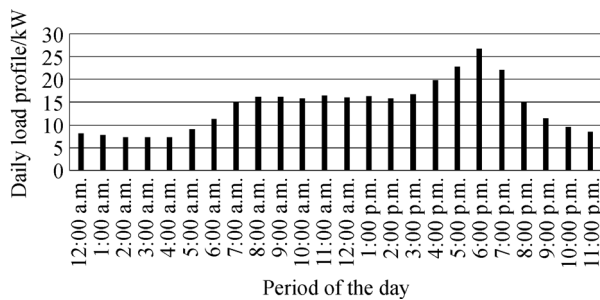
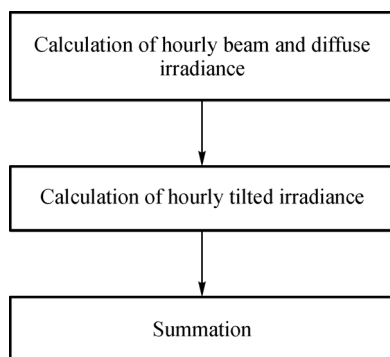
The algorithm can be depicted as a sequence of three basic steps, as illustrated in Fig. 3 [45].

4.3.2 PV array optimization modeling

The PV array optimization modeling was carried out by using the hybrid optimization model for electric renewable (HOMER®) [46]. The power output of the PV array was evaluated using

Table 2 Electricity consumption analysis of a rural community of 200 homes

Description	AC/DC	Intermittent resource-load correlation	Base case load (home)/W	No. of appliance per home/W	Hours of use per day/(h·d ⁻¹)	Days of use per week	Base case load for community/W
TV	AC	Negative	90	1	6	7	18000
Bulb	AC	Negative	18	6	7	7	21600
Fan	AC	Zero	24	3	8	7	14400
Water pump	AC	Positive	Community based	Community based	3	3	20000
Radio	DC	Zero	6	1	5	7	1200
Clinic equipment	AC	Positive	Community based	Community based	5	5	2000
School electronics	AC	Positive	Community based	Community based	5	5	2400
Electricity- daily- AC/(kW·h)							357.256571

**Fig. 2** Daily load profile for design of hybrid energy systems in rural areas of North-west Nigeria**Fig. 3** Flowchart of tilted irradiance calculation

$$P_{PV} = f_{PV} Y_{PV} \frac{I_T}{I_S}, \quad (1)$$

here f_{PV} is the PV derating factor; Y_{PV} , the rated capacity of the PV array (kW); I_T , the global solar radiation incident on the surface of the PV array (kW/m²); and I_S is taken as 1 kW/m². For each hour of the year, HOMER[®] calculates the global solar radiation incident on the PV array by the model of Duffie and Beckmann [43]. The derating factor is a scaling factor that relates the effects of dust to the panel, wire losses, elevated temperature, or anything that could cause the output of the PV array to depart from its expected value under ideal conditions.

4.4 Modeling the wind speed distribution

4.4.1 Wind energy model

The characterization and analysis of the wind speed profile for the site was performed by using the Weibull probability density function [45]. The reason for this is that the Weibull probability density function has been found to conform considerably to the experimental long-term distribution of the wind profile distributions of the site [5,24,47,48].

The two parameter Weibull probability density function is expressed as [49]

$$p(v) = \left(\frac{k}{C}\right) \left(\frac{v}{C}\right)^{k-1} \exp\left[-\left(\frac{v}{C}\right)^k\right], \quad (2)$$

where v is the wind speed (m/s), k and C (m/s) are the two parameters called the shape and scale factors,

$$C = \frac{v}{\Gamma(1 + 1/k)}, \quad (3)$$

where Γ is the gamma function.

A flowchart of the algorithms is shown in Fig. 4 [45].

4.4.2 Energy curve

The energy curve data is the total amount of energy a wind turbine generates over a range of annual average wind speeds, which is expressed as

$$E\bar{v} = 8760 \sum_{x=0}^{25} P_v p(v), \quad (4)$$

where P_v is the turbine power at the wind speed, v and x is the cut-in wind speed (m/s) and 25 is the cut-out wind speed (m/s).

4.5 Modeling the battery bank

The battery bank is modeled as a collection of one or more individual batteries using the HOMER[®] software [50].

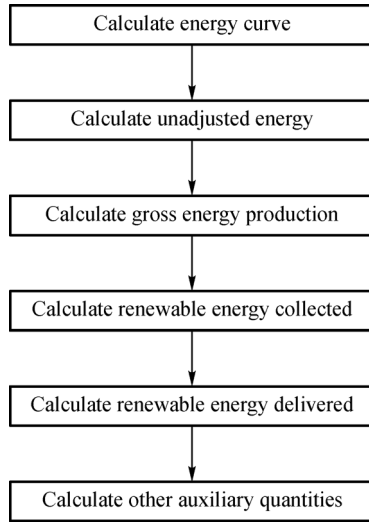


Fig. 4 Flowchart of wind energy model

In calculating the maximum allowable rate of charge or discharge of the battery, HOMER[®] uses the kinetic battery model [51]. This model regards the battery as a two-tank system as symbolized in Fig. 5. In accordance with this battery model, a fraction of the energy storage capacity of the battery is instantaneously available for charging or discharging, while the remainder is chemically bound. The levels in the two tanks are, therefore, tracked by HOMER[®] each hour, while modeling both effects. The number of cycles to failure drops acutely with increasing depth of discharge.

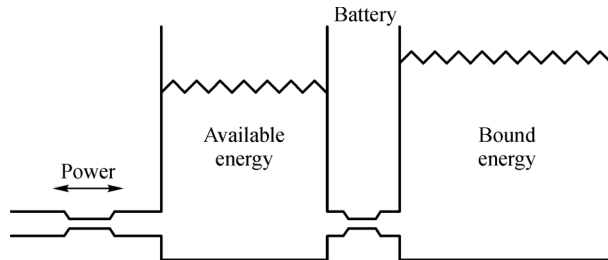


Fig. 5 Kinetic battery model

4.6 Cost benefit analysis

Economics plays a vital role in the selection of the optimal combination of energy resources, as renewable and non-RE sources usually have radically diverse cost characteristics. Renewable sources tilt toward having high initial capital costs and low operating costs, while conventional non-renewable sources usually tend to have low capital and high operating costs. Therefore, in the optimization process, HOMER[®] operates the system in a way as to diminish the total net present cost (NPC).

The life-cycle cost (or NPC) analysis comprises all costs

incurred within the system life span. These include the costs of initial construction, component replacements, maintenance, fuel, cost of buying power from the grid, and miscellaneous costs such as fines ensuing from pollutant emissions. Revenues take into account the income from selling power to the grid, plus any salvage value that occurs at the end of the project lifetime. As for the NPC estimation, costs are seen as positive and revenues are negative. The NPC is equal to the NPV, but with an opposing sign, i.e., negative NPC equals a positive NPV-signifying value.

HOMER[®] assumes that all prices escalate at the same rate over the project lifetime. Hence, the real interest rate, which is roughly equal to the nominal interest rate minus the inflation rate, serves as an input into the software.

For each component, the capital, replacement, maintenance, and fuel costs, along with the salvage value and other costs or revenues, add up to the annualized cost of the component. The annualized costs are then summed for each component, together with any miscellaneous costs, so as to find the total annualized cost of the system. The total net present cost is

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, R_{proj})}, \quad (5)$$

where $C_{ann,tot}$ = total annualized cost; i , the annual real interest rate (the discount rate); R_{proj} , the project lifetime; and $CRF(\bullet)$, the capital recovery factor, given by

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1}, \quad (6)$$

where i is the annual real interest rate and N is the number of years.

The levelized cost of energy (LCOE) is

$$LCOE = \frac{C_{ann,tot}}{E_{prim} + E_{def} + E_{grid,sales}}, \quad (7)$$

where $C_{ann,tot}$ is the total annualized cost, E_{prim} and E_{def} are the total amounts of primary and deferrable load, respectively, that the system serves per year, and $E_{grid,sales}$ is the amount of energy sold to the grid per year.

4.7 Specifications of wind turbines and solar panel

Two Portland general electric (PGE) 20/25 wind turbines [50] were cumulatively employed for this study, each being specified with a cut-in speed of 3.5 m/s, a low wind cut-out speed of 1.7 m/s, a high wind cut-out speed of 25 m/s, a rated wind speed of 9 m/s and a rated power at a rated wind speed of 25 kW. The available hub heights for this turbine are 24 m, 30 m and 36 m, while the rotor diameter is 20 m.

The solar cell specification was that of a 1 kW Sun power mono-crystalline silicon cell [42] with a collector area of 5.1 m², an efficiency of 19.6%, a nominal operating

cell temperature of 45°C and a temperature coefficient of 0.40%/°C. Miscellaneous losses were taken as 10% while the array slope angle utilized was the latitude angle of the site. Therefore, as required by the load, the size of the solar panel was increased to suite the load demand, while the solar collector area increases with other parameters remaining constant. Table 3 shows the cost of the components used in the design of the HES. The installation costs are embedded in each component cost.

5 Results and discussion

Figures 6 and 7 present the average monthly and annual solar radiation profiles from 1987 to 2010. It can be observed that the monthly average solar radiation of the 24 years ranged from 4.132 (kW·h/(m²·d)) in August to 6.117 (kW·h/(m²·d)) in March, while the yearly average ranged from 5.1 (kW·h/(m²·d)) in 2005 to 5.55 (kW·h/(m²·d)) in 1987.

In Figs. 6 and 7, more variability was found to be associated with the annual average than those of monthly average data. Moreover, the fact that the hours equaled or exceeded for a range of mean measured solar radiation (Fig. 8) across the period revealed that the PV array generates power output for only about 4358 h of the 8760 h in a year. This corresponds to about 49.7% of the hourly duration in a whole year. This, however, is caused by the fact that solar radiation, unlike wind speed, occurs only during the daytime and Sokoto has a 20 year average sunshine daily duration of about 7.7776 h. Figure 7 also demonstrates that at half the duration of power production of about 2179 h, an electricity production of about 49 kW is sustained. This value equates to about 107% of primary

peak — load demand required in the rural community. The remaining 20 kW deferrable load of 16 kW is met at the period of lowest demand at noon (see Fig. 2). The data values, therefore, indicate that the site can sufficiently harness the immense solar energy potential of Sokoto and provide solar electricity to the rural poor.

The correlation of the monthly average solar radiation and electricity generated using a 135 kW PV module is presented in Fig. 9 which reveals a good correlation between the incident irradiation and the electricity generated, and demonstrates that the 24 years monthly average solar radiation ranged from 4.132 (kW·h/(m²·d)) in August to 6.117 (kW·h/(m²·d)) in March.

Potentially, the monthly average electricity production ranged from 19.316 kW in August to 30.071 kW in November. The difference in the months of maximum electricity production and the maximum solar radiation measurement can be associated partly with the fact that the values are averaged over the days of the months. It can also result from the fact that the simulation takes into consideration the influence of other parameters such as the average daily ambient temperature as well as variables such as air pressure, relative humidity and wind speeds. These factors affect the magnitude of the solar electricity generated from the panels and their values vary across the months. The degree of difference, however, differs minimally because of the overriding influence of daily global solar radiation. Table 4 presents the results of matching production with consumption of the solar electricity.

From Table 4, it is observed that the most cost effective system design for the PV standalone system with a loss of load probability (LOLP) of 0.01 [52–54], gave an excess electricity equivalent to 35% of annual generation. This

Table 3 Cost of components used in the design of HES (installation cost embedded in component cost)

Component	Interest rate/%	Project life time	Cost/(\$·kW ⁻¹)	O & M/\$	Replacement cost/(\$·kW ⁻¹)
Wind turbine	6	20 years	1800	400/year	1800
Solar panel	6	25 years	3000	0/year	1500
Battery	6	10 years (float)	100	20/year	100
Converter	6	12 years	500	80/year	500
Diesel generator	6	15000 h	350	0.050/h	300

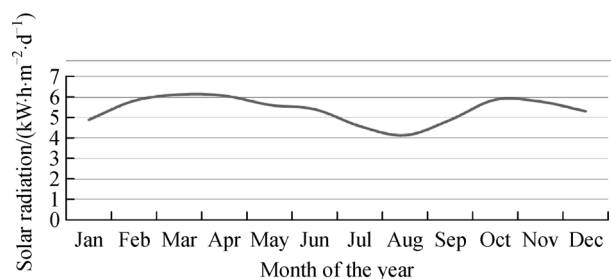


Fig. 6 Monthly average radiation in Sokoto

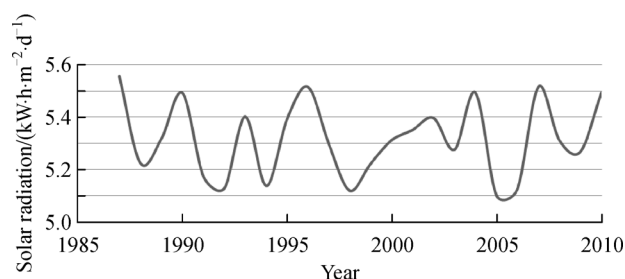


Fig. 7 Annual average solar radiation in Sokoto

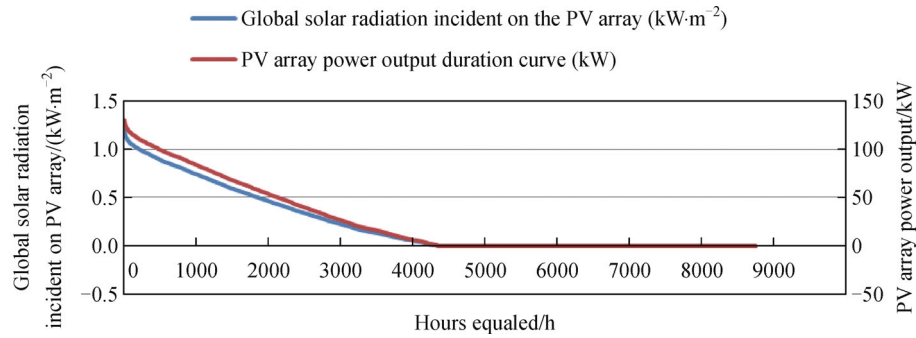


Fig. 8 Plot of annual average hours equaled or exceeded for different radiation and PV power outputs of 24 years in Sokoto

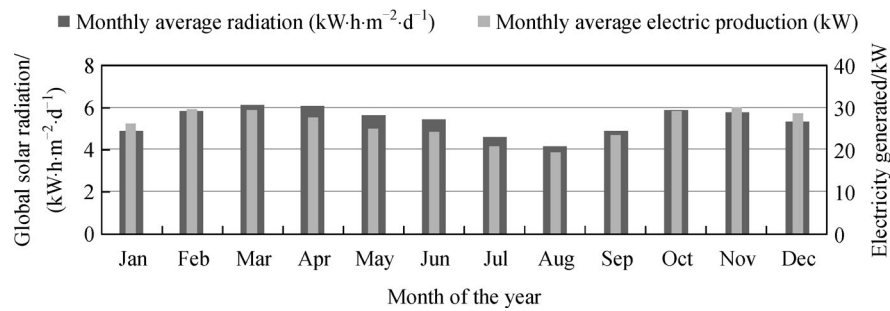


Fig. 9 Correlation between monthly average solar radiation and electricity generated via a 135 kW solar panel in Sokoto

Table 4 Electricity consumed as a percentage of PV and wind standalone production

Quantity	PV energy value/(kW·h·a ⁻¹)	Wind energy value/(kW·h·a ⁻¹)
PV array/ Wind turbine production	228102	293689
load met (LOLP of 0.01)	129824	129701
Excess electricity	72112	154793
Unmet electric load	830	957

was caused by the fact that the power generated via solar energy is not constant throughout a 24-h day. The 24-year average sunshine hour for Sokoto is 7.7776 h. However, there were days, months and the rainy periods in which the average sunshine duration were between 3 h and 4 h. Thus a design that will serve a load profile of 200 rural homes including the requirement for battery charging will of necessity take into consideration the days of limited solar radiations. The battery profile will be such that will have 55.7 h of autonomy and an initial state of charge of 50% (to extend battery life [55–57]). Thence, with a design covering an entire year, it gives rise to an excess in the energy generated annually when the period of higher sunshine duration is balanced with those of lower duration. The excess may be utilized in the form of embedded generation, as indicated in the policy document of the Nigerian Electricity Regulatory Commission (NERC). Embedded generation in this context is defined as the generation of electricity that is directly connected to and

evacuated through a distribution system which is connected to a transmission network operated by a system operations licensee. It can be described as a form of generation where excess RE generated by a consumer ranging from 1000 kW to 5000 kW can be sold to a nearby distribution network [58,59]. This has the advantage of a reduction in the LCOE, as Eq. (7) indicates that an increase in grid sales serves to reduce the LCOE. On the other hand, the excess may not be eliminated (or sold to the grid) but stored when less than 1000 kW. The battery specification employed for the study was that with a nominal capacity of 1188 kW·h and a usable capacity of 832 kW·h. The optimized nominal capacity (or rated capacity) of the battery is the amount of energy that could be withdrawn from it at a particular constant current, starting from a fully charged state. Moreover, the usable capacity is lower because the rated capacity is reduced by a yearly storage depletion whose magnitude depends on the frequency of cycling energy in and out of the battery as well as its

maximum depth of discharge. The rated capacity is also reduced by the yearly losses in the battery. For this PV standalone system, its usable capacity is 70% of its nominal capacity. The expected life of the battery is 8.05 years from the design, though the float life for this Trojan battery is specified as 10 years. This is caused by the storage depletion of 0.22% per year, such that 19.5% of the battery life is lost over its life time. Also the energy input to the battery is reduced by 14.6% losses and storage depletion of 0.22%, leading to an energy output of 14.8% less than the energy input into the battery on a yearly basis. The monthly state of battery charge (SOC) is presented in Fig. 10, according to which the average monthly value of the battery state of charge by percentage declines between June and September, corresponding to the same decline in average monthly solar radiation profile (see Figs. 6 and 9). This proves the variation in the average monthly electricity production (see Fig. 9).

The NPC (Fig. 11) represents the life cycle cost, which captures all the cost throughout the operational life (25 years) of the system. It gives a salvage value of about 4.56% of the total NPC for the system. It must be noted that a project life of 25 years was chosen because of the life span of solar panels, but this was found to be expensive,

because its LCOE is \$0.458/(kW·h). When the project lifespan was extended to 100 years, a LCOE of \$0.426/(kW·h) was obtained. This was brought about by the inclusion of the replacement cost for each component in the analysis, thereby making the design go beyond the required 20 five years' lifespan of the module. In addition, since each component cost is expected to reduce over the years, the LCOE is projected to decline further. The analysis reveals that 52% of the total NPC was the cost related to the solar panel, while 41% is related to initial, maintenance and replacement costs of the battery. The converter bears the remaining 6% of the total NPC. This further reveals that with the present rate of decline in prices of solar panels [60–62], the effect on life cycle cost will progressively decline, making PV systems much more competitive with grid electricity.

5.1 Prospect of standalone wind-to-electricity project

The results of wind profile analysis at the site are as shown in Figs. 12 and 13. These two figures demonstrate that the monthly average wind speeds of the 24 years ranged from 4.8952 m/s in October to 8.9238 m/s in June, while the yearly average ranged from 5.692 in 1993 to 8.62 m/s in

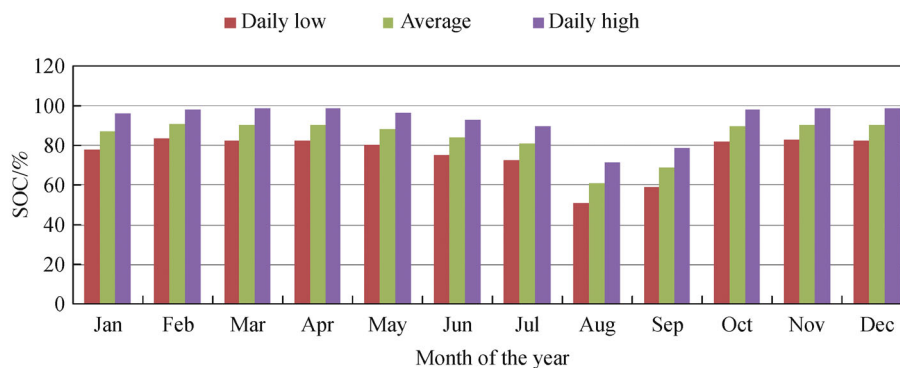


Fig. 10 Monthly battery state of charge (SOC) for PV standalone system

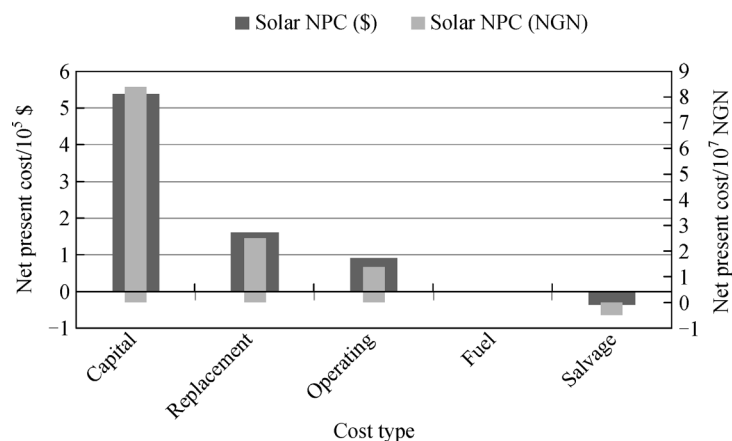


Fig. 11 NPC by cost type for PV standalone system in Sokoto

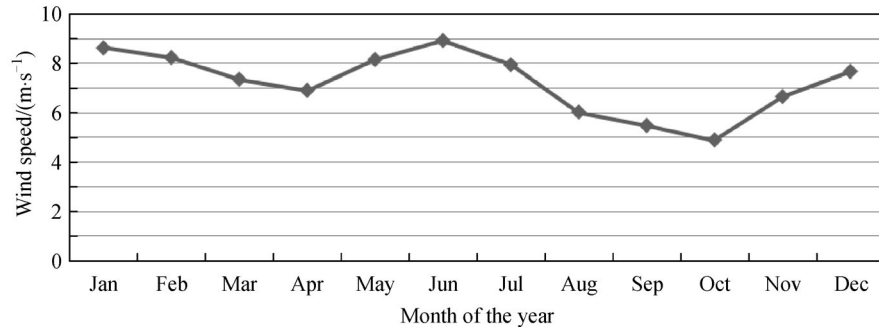


Fig. 12 Plot of monthly average wind speeds of 24 years

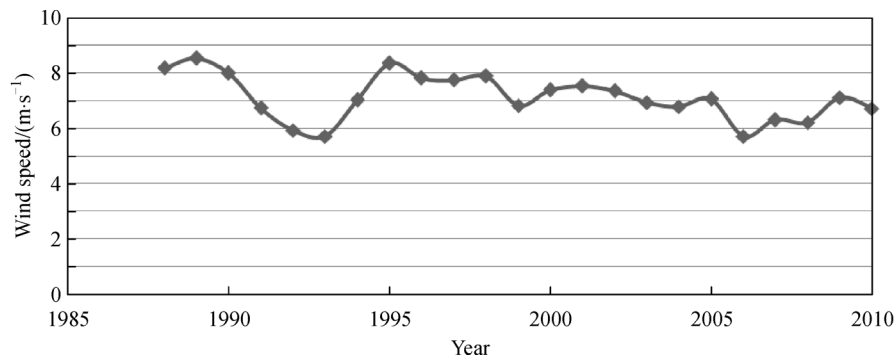


Fig. 13 Plot of annual average wind speeds of 24 years

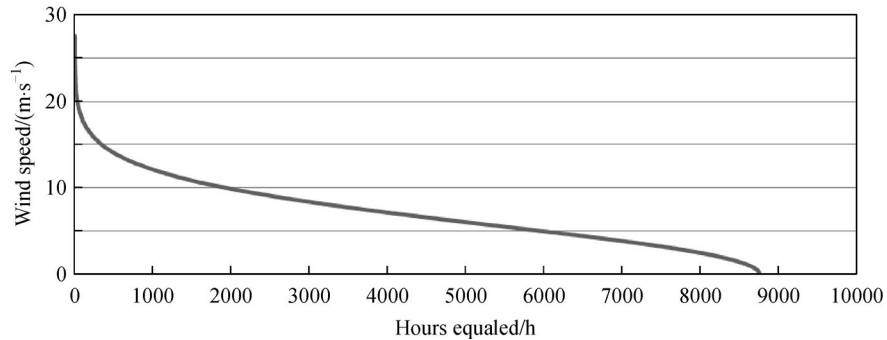


Fig. 14 Plot of annual average hours equaled or exceeded for different wind speeds of 24 years

1987. Moreover, the fact that the hours equaled or exceeded for a range of mean measured wind speeds (Fig. 14) across the period revealed that 87.5% of the data spread are values above 3.0 m/s. Thus the site is compatible with modern wind turbines for power generation throughout the year.

Figure 15 reveals a good correlation between the wind speed profile and the electricity generated. The monthly average electricity produced ranged from 22.617 kW in October to 38.554 kW in June. This clearly indicates that unlike the energy generated from the PV system, wind energy is mainly correlated to the wind speed profile of the

site without much sensitivity to other meteorological variables such as air temperature and relative humidity. The relationship between the magnitude of potential wind-electricity production and consumption is presented in Table 4.

Table 4 reveals an excess electricity equivalent to 54% of annual generation. The reason for this is that wind power is generated for about 88% of the time (see Fig. 14). Therefore, since the site has the ability of generation almost every hour of the day, an optimal battery size of 21.8 h of autonomy can be employed. Besides, since the rated speed for the two PGE 20/25 turbines used in the

design is 9 m/s, Fig. 14 shows that at about 30% of annual hourly duration, the turbines can produce at the rated capacity. This, therefore, gives good returns on investment and an opportunity for embedded generation [58,59].

Table 5 reveals that the WSS has only 39% of both rated capacity, and the usable capacity of the battery requirement for the PV standalone system (PSS), equivalent to a savings of 61% of battery cost for the WSS, hence making the WSS more cost effective than the PSS, since the battery has an overwhelming cost implication on both systems (see Table 3). The results of the analysis of the monthly battery SOC are presented in Fig. 15.

Figure 16 demonstrates that the average monthly value of the battery state of charge by percentage never drains up to the 50% minimum [55–57]. The lowest value of about 65% was obtained for October. Thus, from the state of charge usage frequency for the WSS, the battery remains 95%–100% charged for about 70% of the year. Therefore, the expected life of the battery is equal to its float life of 10 years as specified by the manufacturer. This is because the

wear cost losses are very low as the batteries hardly are cycled below 95% of its energy capacity. The storage depletion as seen from Table 5 was found to be approximately 0% with 15% losses experienced annually for the WSS.

Figure 17 presents the NPC of employing only the WSS for power generation at the community. Comparing the NPC for the WSS and PSS reveals differences in the value for both systems. This results from the fact that the cost of wind turbine per kW is less than that of solar panels (see Table 3), and, additionally, the wind energy resource at the location is very close to the turbines rated speed. This, therefore, enabled production at its power rating. This was the reason why a larger capacity rating of 135 kW was required for the solar panel as compared to 50 kW (two 25 kW modules) for the wind turbines. The total NPC was found to be 64% less for the WSS than the total NPC for the PSS. It was also observed that the greatest differential of NPC by cost type is with the capital cost. The capital cost for the WSS is 72.5% less than that of the PSS.

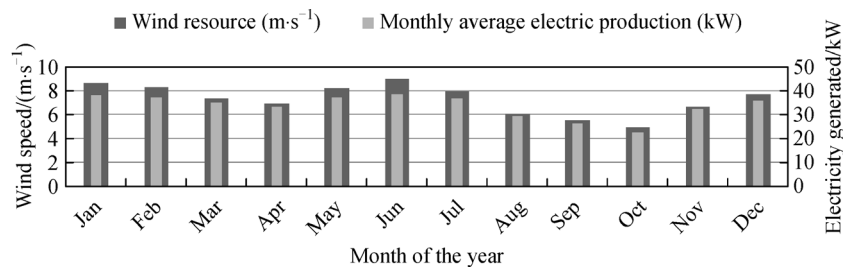


Fig. 15 Correlation between monthly wind speed and electricity generated via two PGE 20/25 50kW turbines

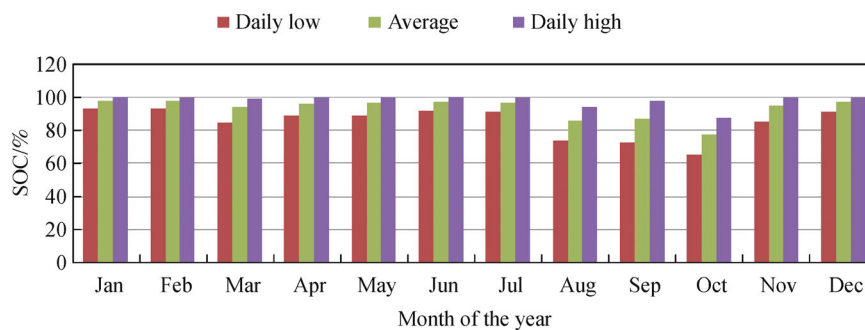


Fig. 16 Monthly battery state of charge for WSS

Table 5 Trojan L16P battery specification—comparison between wind and PV standalone systems

Quantity	Wind battery capacity/(kW·h·a ⁻¹)	PV battery capacity/(kW·h·a ⁻¹)
Energy in	22347	79444
Energy out	18995	67703
Storage depletion	0	175
Losses	3352	11566

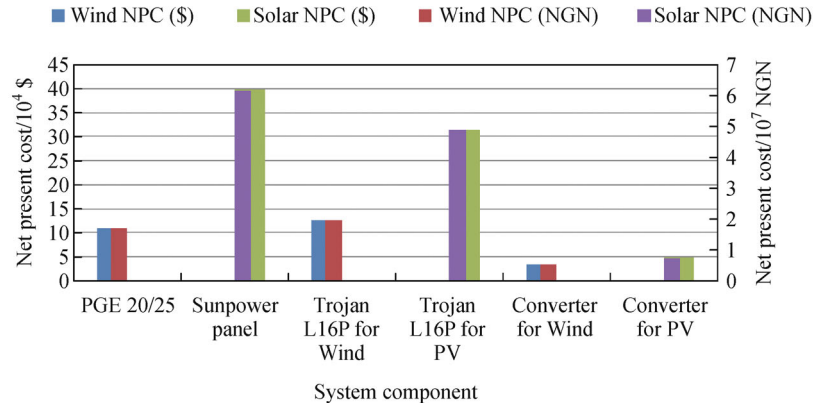


Fig. 17 NPC summary—comparison between wind and PV standalone systems

Moreover, the cost of replacement, operating cost, and the salvage value were lower by about 53%, 36%, and 38%, respectively.

5.2 Analysis of the potential of solar-wind hybrid system

The idea of hybridizing RE resources is that the base load is to be covered by the most abundant and firmly available renewable source. The other irregular source would supplement the base load to cover the peak load of the off grid or isolated grid system.

In practice, the use of HESs is also a workable way to realizing trade-off solutions in terms of costs. Therefore, in order to suit the load demand, HESs are implemented to combine solar and wind energy units, thereby diminishing power fluctuations. The combination of these RE systems will then definitely reduce the technical requirements and by extension, the cost of the storage batteries. Table 6 and Fig. 18 present the economic cost of utilizing the standalone wind, solar and diesel systems for power generation whether singly or as hybrid systems.

Table 6 shows the best renewable technology for the community based on NPC. The wind-diesel though has a better LCOE (Table 6) than wind-PV (which has a better total NPC (Table 6)). This is, however, caused by the fact that the wind-PV production efficiency is 23% lower than

the wind-diesel annually. Table 6 lists the ranking of the standalone systems by LCOE. The results in Table 6, however, preclude the possibility of grid connection, and as a result, the cost of selling excess generation is not taken into account. Therefore, the PV-wind hybrid has a better NPC than the wind-diesel since the excess electricity produced serves as revenue which lowers the overall cost.

Based on the results, it is evident that the solar resource, though very much feasible for this site, falls below the potential of wind energy with LCOE difference of \$ 0.31. This means that there is a 75% savings per kilowatt-hour if WSS was used instead of the conventional diesel standalone system (DSS). Further to this, the result yielded a 26% savings per kilowatt-hour because PSS was used instead of DSS.

Furthermore, Table 6 shows that the solar-wind hybrid did not provide a reasonable trade off in terms of cost when compared with wind alone and wind-diesel hybrid. It is, however, more economical than employing either only solar, diesel or solar-diesel hybrid. Therefore, the best renewable technology that fulfills all the technical requirements, as well as being the most economically viable alternative for power generation at the rural community of 200 homes in Sokoto, North-West Nigeria, is the WSS. With the present government's reform of electric tariff regime ongoing in Nigeria, grid electricity

Table 6 Results of econometrics analysis for the deployment of various power generating RE technologies at the rural community (ranking by total NPC)

Technology	Total NPC/\$	Total NPC/NGN	Initial capital/\$	Initial capital/NGN	LCOE/ (\$·kW ⁻¹ ·h ⁻¹)	LCOE/(NGN·kW ⁻¹ ·h ⁻¹)
Wind	8527	1321685	145700	22583500	0.15	23.87
Wind-diesel	13177	2042435	93150	14438250	0.16	24.34
Wind-PV	11254	1744370	290090	44963950	0.26	40.30
PV-diesel	25174	3901970	302900	46949500	0.38	58.28
PV	17147	2657785	541300	83901500	0.46	70.99
Diesel	79866	12379230	12250	1898750	0.62	95.95

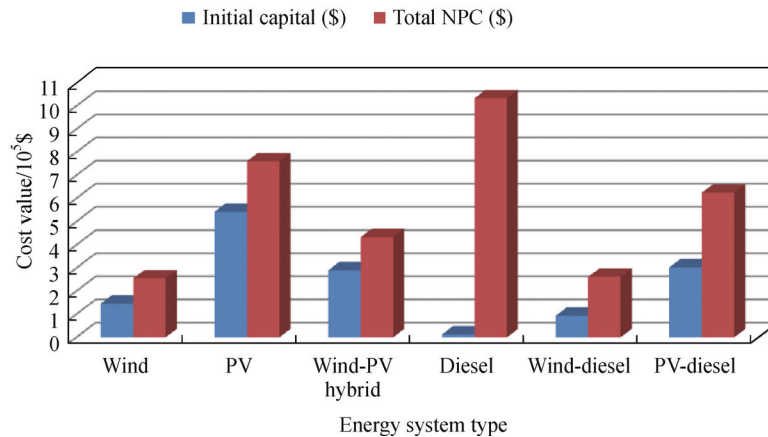


Fig. 18 Comparison between initial capital and total NPC

prices have been on the increase [63]. Besides, based on the fact that research is ongoing to lower the price of wind turbine materials and solar panels, the competitiveness of RE generation will be on the increase.

Together with the WSS and PSS, the wind-solar hybrid contributes 100% of RE. The respective contributions of all hybrid systems were 67%–33% (wind-solar), 89%–11% (wind-diesel), and 84%–16% (solar-diesel). These points to an environmental savings of 279 tons of CO₂ greenhouse gas emission (GHG) annually. This reveals not only the immense advantage of producing energy from RE beyond just the cost implication alone, but also the possibility of having a cleaner and safer environment.

5.3 Government efforts toward RE utilization

A national energy policy was approved by the Federal Government of Nigeria in 2003 with the overall thrust of optimal exploitation of the nation's energy resources. Some policy documents, which include the RE master plan, have also been generated. The major focus of the government is to develop the nation's energy frontiers to mix both conventional and nonconventional energy sources to cater for the growing population [64–66]. One of the major highlight of the RE master plan is to meet 20% of the nation's electricity needs using RE sources by 2020. Based on this, the government has pushed forward some of its plans through the recent deregulation of electricity generation and distribution to pave the way for states and local councils to generate and distribute electricity within their areas [67,68]. This, thereby, opens opportunity for adoption of standalone energy generation facility in rural areas of Nigeria. In addition, there is the multi-year tariff order (MYTO) for generation which presents a 5-year plan (2012 – 2017). It stipulated a feed-in tariff that provides a very good and competitive pricing for potential private investors [59]. Table 7 provides the cost benefit accruable to an investor, when provision is made to generate

standalone renewable electricity for the studied rural community of 200 homes with the option of selling excess generation to the grid.

The negative sign indicates value rather than cost, hence the negative NPC becomes the NPV of the system and the negative LCOE indicates that both systems actually yield profit in \$(kW·h) rather than a cost of producing energy.

The Federal Government of Nigeria has in its Renewable Electricity Policy Guidelines [65] shown willingness to support RE projects financially, especially those for remote and rural communities, through its Renewable Electricity Trust Fund (RETF). The move intends to promote, support and provide renewable electricity through private and public sector participation. The financing options include monies appropriated by the National Assembly, equity investments, debt financing, grants, micro credit and donations, gifts and loans from all eligible local and international sources. Therefore, willing investors can take advantage of the provision by investing in renewable electricity generation within rural communities of Nigeria.

6 Conclusions

An assessment and design of PV and wind energy power systems in standalone and hybrid format for a local meteorological site in Sokoto, North-west Nigeria has been conducted. The conclusions from this study are as follows:

- 1) The technical and econometric requirement for the PSS and WSS were found to be a 135 kW solar panel (mono-Si technology), a 1118 kW·h Trojan battery capacity and a 35 kW converter, (with an initial capital of \$541300, a total NPC of \$17147 and an LCOE of \$0.46/(kW·h)), for a solar standalone system; Two 25 kW wind turbines, a 464 kW·h Trojan battery capacity, and a 35 kW converter, (with an initial capital of \$145700, a total NPC of \$8527 and an LCOE of \$0.15/(kW·h)) for a WSS.

Table 7 Investor profit on embedded generation for a 5-year Project Life Span with the present MYTO for Sokoto, Nigeria

Technology	Turbine/PV panel rating /kW	Initial capital/\$	Total NPC	Annualized cost /(\$·a ⁻¹)	LCOE /(\$·kW ⁻¹ ·h ⁻¹)	Profit/(\$·a ⁻¹)
Wind embedded generation	3000	5438300	−8137331	−1931774	−0.14	2084025
PV embedded generation	7500	25045 200	−3483146	−827655	−0.074	814592

2) The WSS was found to be the optimal means of producing renewable electricity in terms of NPC as well as LCOE at an LCOE of \$0.15/(kW·h) (NGN 23.87/(kW·h)). This is quite competitive even with grid electricity, at a present cost of about \$0.09/(kW·h) (NGN 13.45/(kW·h)) and approximately 410% better than the conventional DSS at an LCOE of \$0.62/(kW·h). The WSS should then become a priority for use in wind electric pumping systems because it is ideally suited for irrigation and other water pumping activities required in farm settlements that are not connected to the national grid. This is coupled with the fact that most rural communities in Northern Nigeria are farm settlements. Besides, the presence of a favorable wind profile for this site means that wind power can be competitive with conventional sources of power production. Thus wind farms in excess of 100 MW can be established in the studied area with the wind farms being grid-connected or forming the backbone of a mini grid for remote areas far away from the main grid.

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